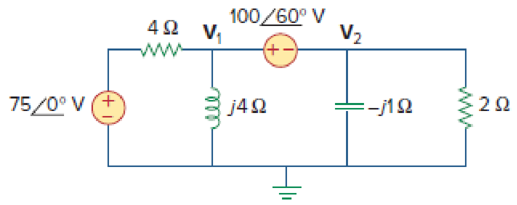
Problem

Calculate **V**1 and **V**2 in the circuit shown in Fig. 10.6.

Figure 10.6:



Step-by-step solution

Step 1 of 3

Consider the following circuit diagram:

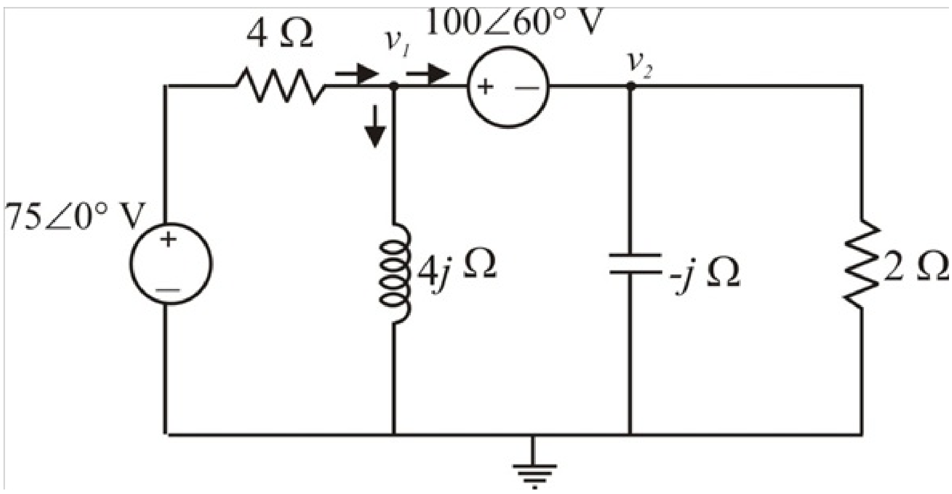


Figure 1

Step 2 of 3

Apply Kirchhoff's current law at node 1 of circuit in Figure 1.

$$\frac{\mathbf{V}_1 - 75\angle 0^\circ \text{ V}}{4\ \Omega} + \frac{\mathbf{V}_1}{j4\ \Omega} + \frac{\mathbf{V}_2}{-j1\ \Omega} + \frac{\mathbf{V}_2}{2\ \Omega} = 0$$

$$\mathbf{V}_1 - 75\angle 0^\circ - j\mathbf{V}_1 + j4\mathbf{V}_2 + 2\mathbf{V}_2 = 0$$

$$\mathbf{V}_1(1-j) + \mathbf{V}_2(2+j4) = 75 \quad \text{...... (1)}$$

From Figure 1, write the following expression:

$$\begin{aligned}\mathbf{V}_1 &= \mathbf{V}_2 + 100\angle 60^\circ \\ &= \mathbf{V}_2 + (50 + j86.60)\end{aligned}$$

Substitute the value of $\mathbf{V}_1$ in equation (1).

$$\begin{aligned}75 &= (\mathbf{V}_2 + 50 + j86.60)(1-j) + \mathbf{V}_2(2+j4) \\ &= \mathbf{V}_2 - j\mathbf{V}_2 + (50 + j86.60) - j(50 + j86.60) + 2\mathbf{V}_2 + j4\mathbf{V}_2\end{aligned}$$

$$\begin{aligned}\mathbf{V}_2 &= \frac{-61.60 - j36.60}{3 + j3} \\ &= 16.88\angle 165.72^\circ \text{ V}\end{aligned}$$

Therefore, the value of voltage $\mathbf{V}_2$ is $\boxed{16.88\angle 165.72^\circ \text{ V}}$.

Step 3 of 3

Calculate the value of $\mathbf{V}_1$.

$$\begin{aligned}\mathbf{V}_1 &= \mathbf{V}_2 + 100\angle 60^\circ \\ &= 16.88\angle 165.72^\circ + 100\angle 60^\circ \\ &= 96.8\angle 69.66^\circ \text{ V}\end{aligned}$$

Therefore, the value of voltage $\mathbf{V}_1$ is $\boxed{96.8\angle 69.66^\circ \text{ V}}$.

Therefore, the voltages $\mathbf{V}_1$ and $\mathbf{V}_2$ values are calculated.